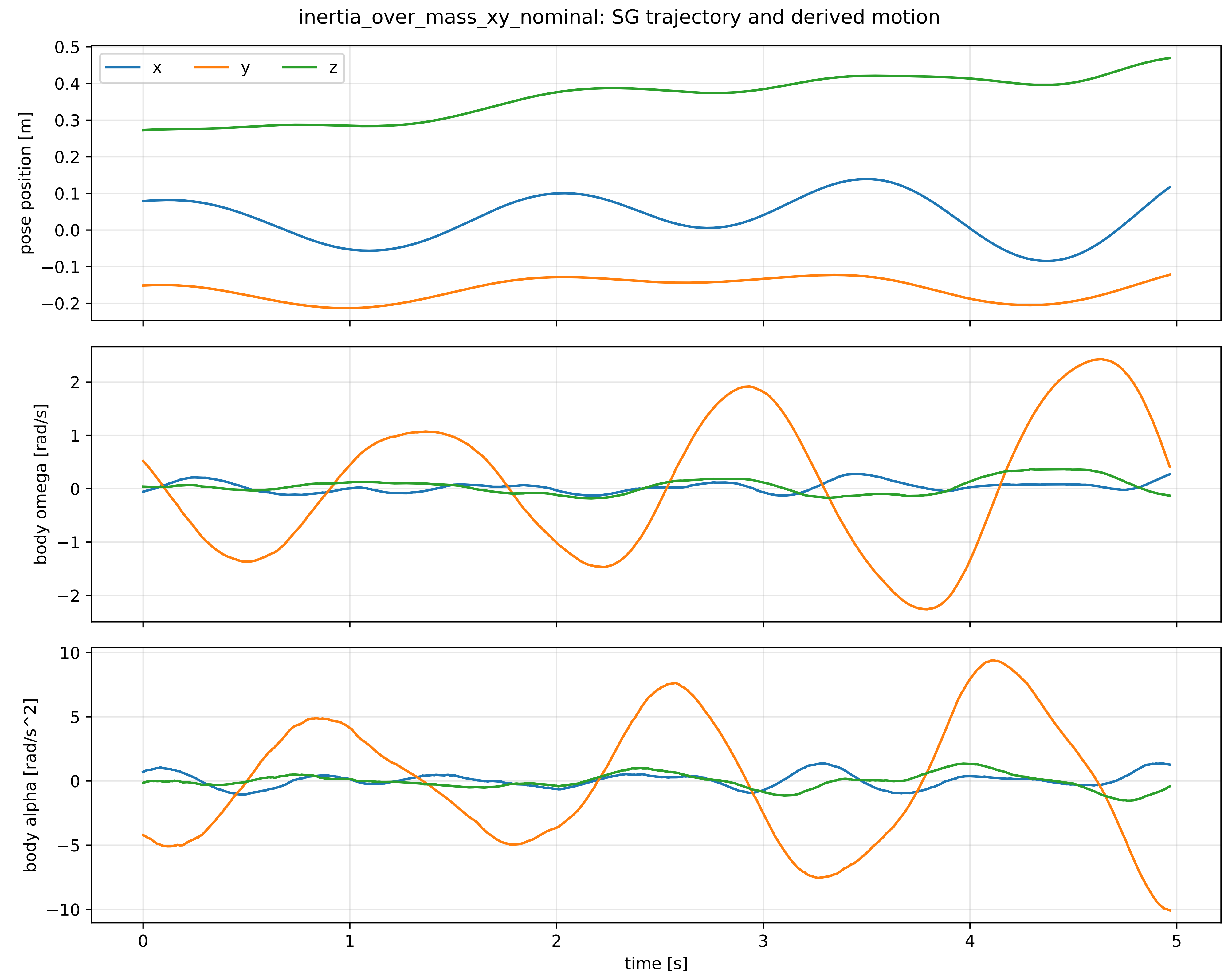
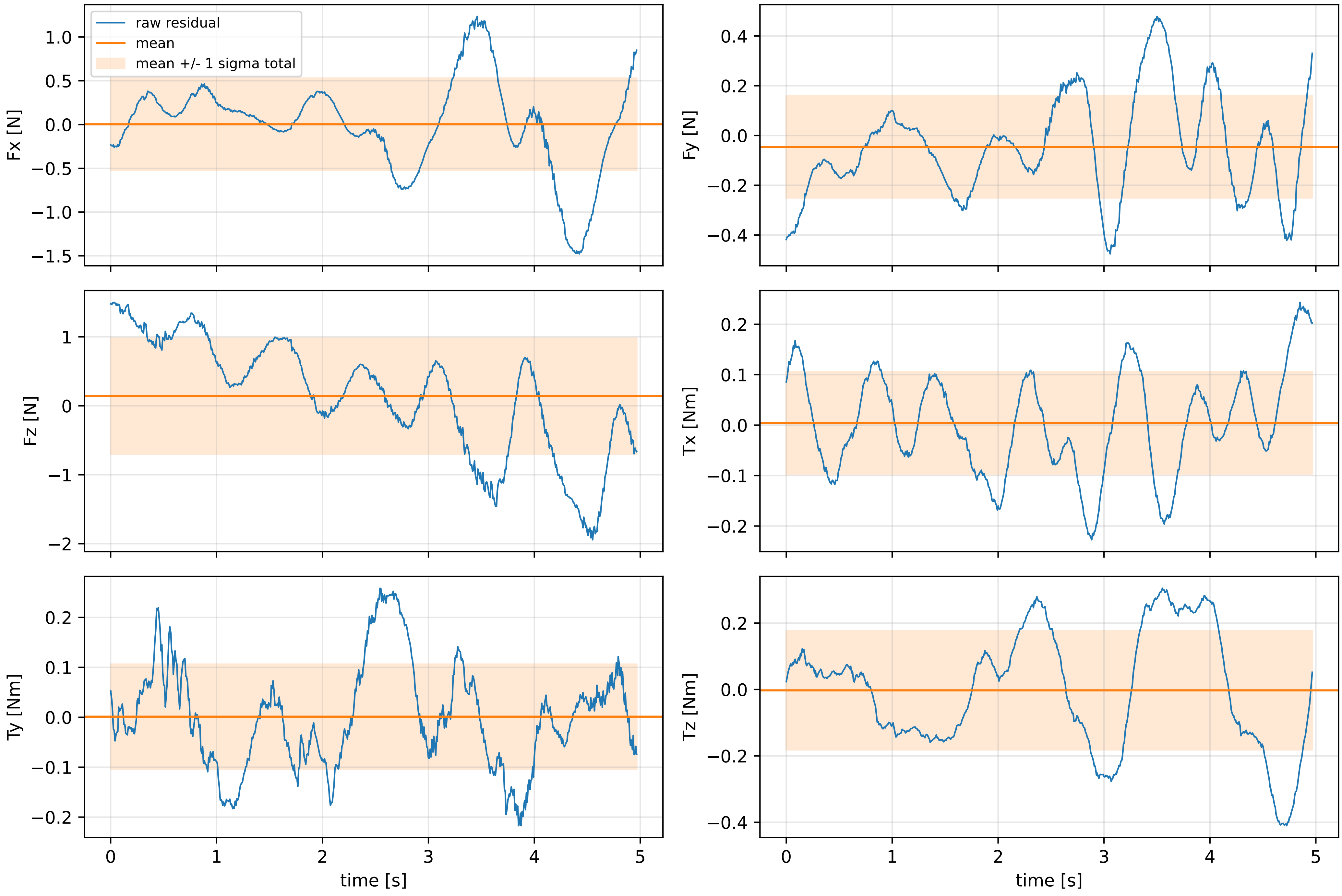
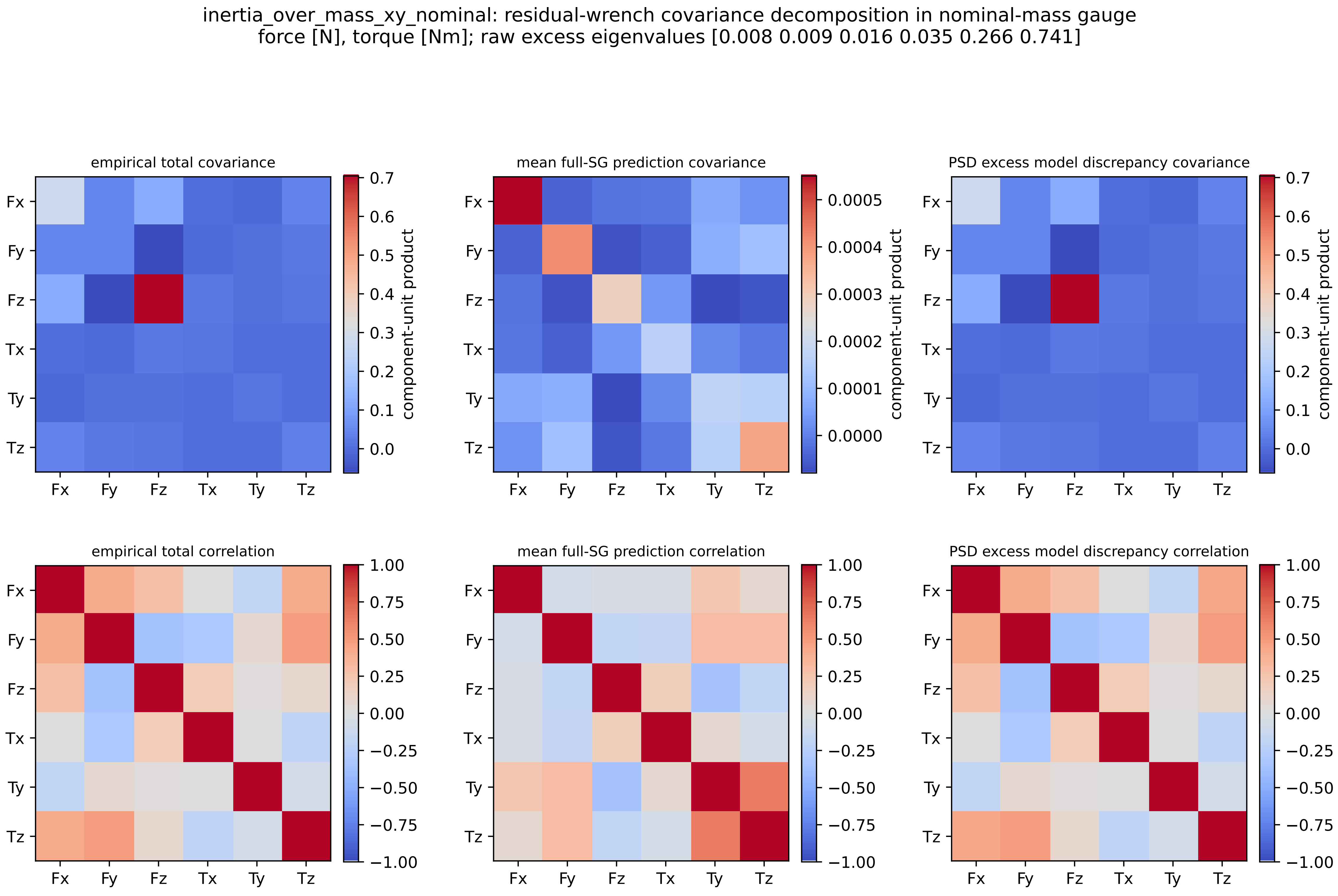


inertia_over_mass_xy_nominal: SG trajectory and derived motion
x
y
z
pose position [m]
body omega [rad/s]
body alpha [rad/s^2]
time [s]

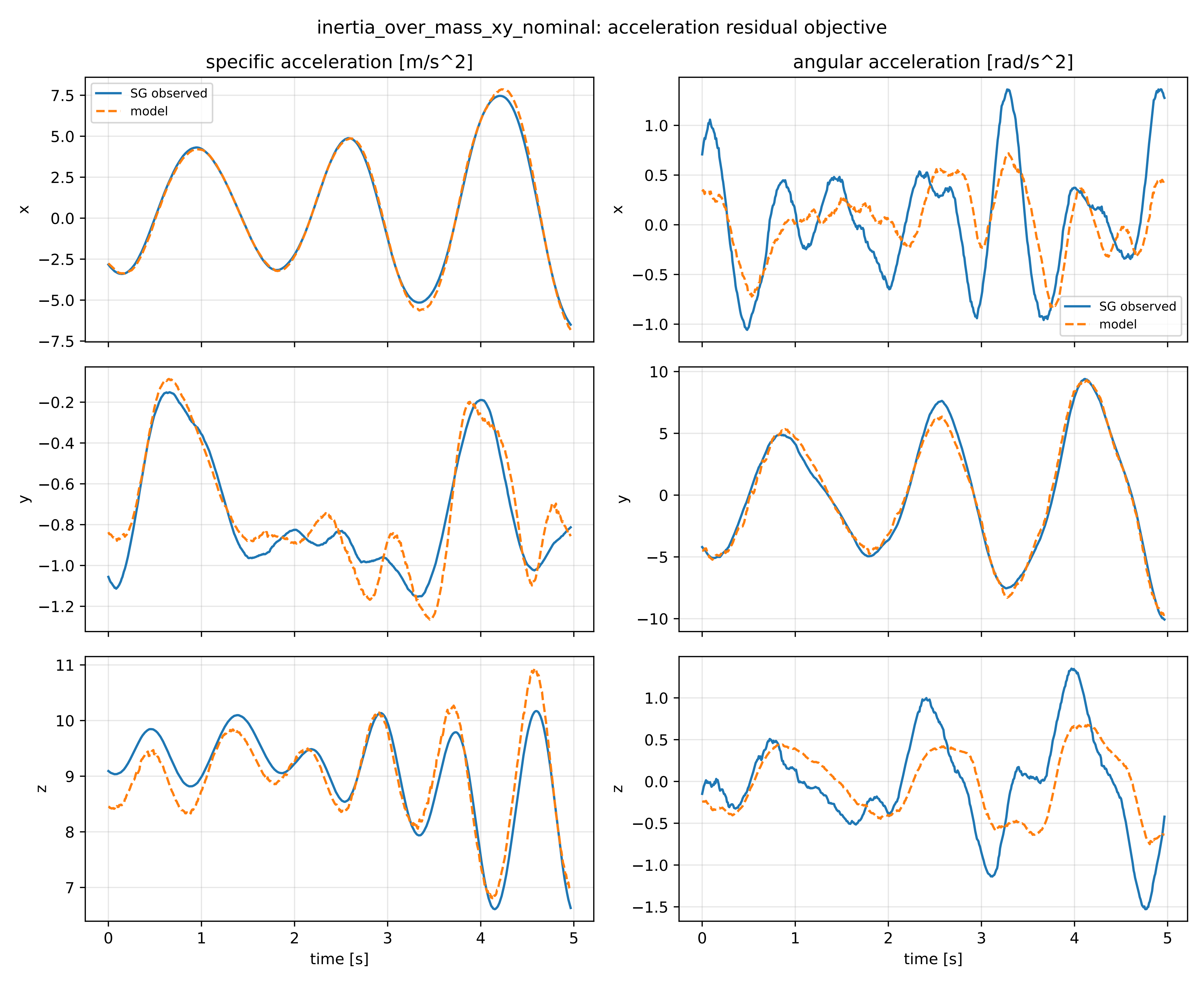
inertia_over_mass_xy_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg

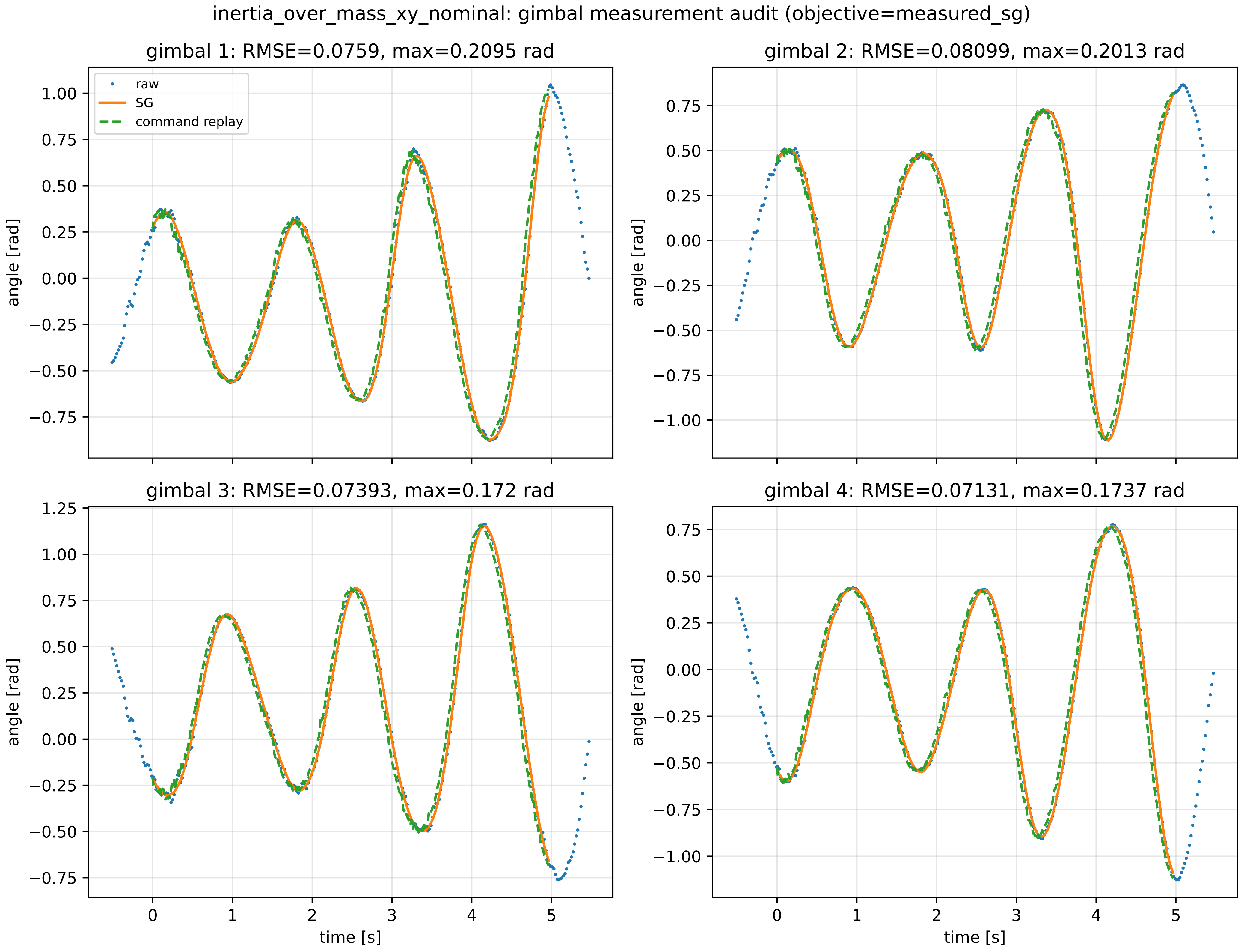


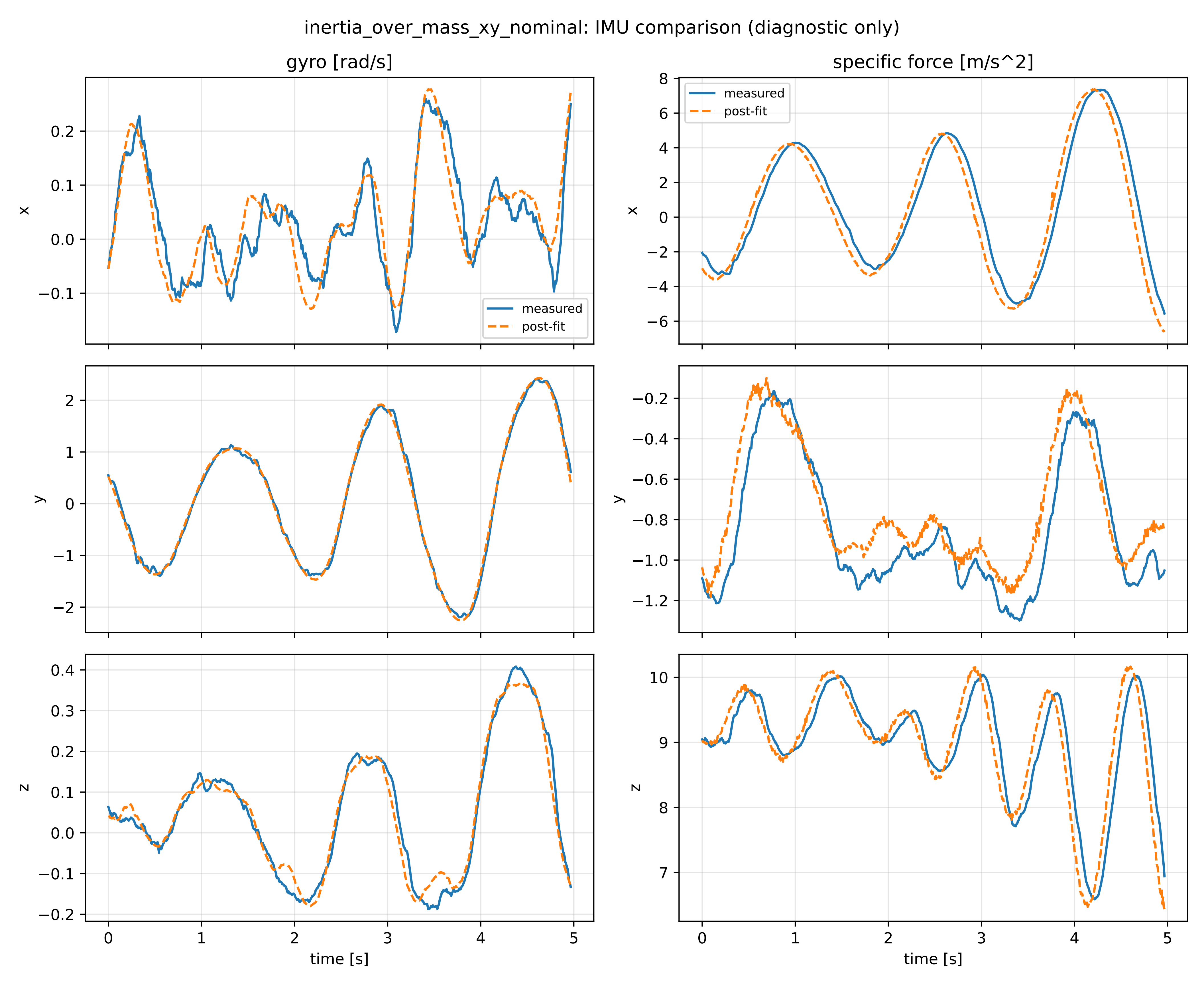
vector RMS about zero: force 1.0215 N, torque 0.23069 Nm; component std about mean: force [0.5258 0.2043 0.8394], torque [0.1016 0.1049 0.1788]



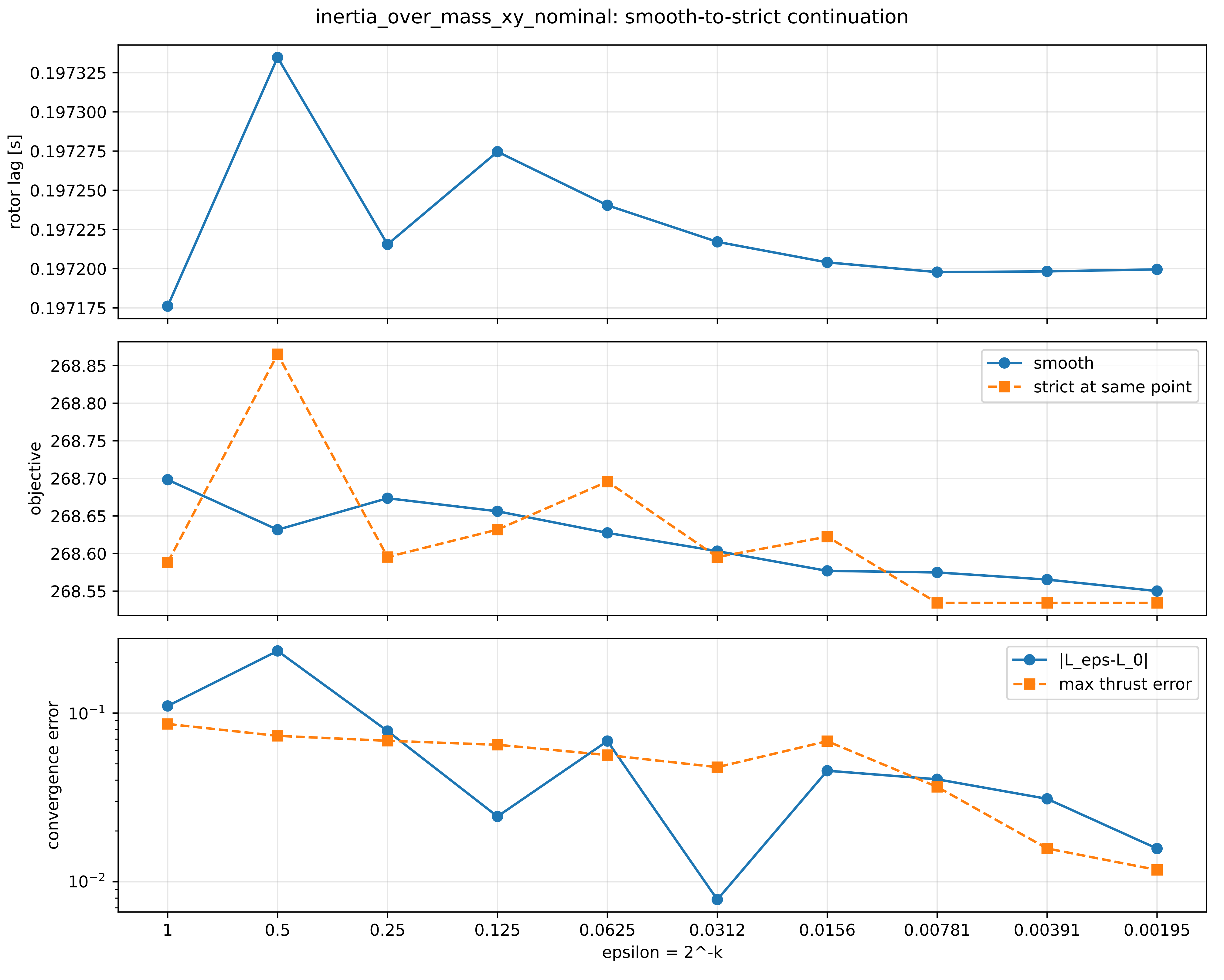
inertia_over_mass_xy_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.008 0.009 0.016 0.035 0.266 0.741]
empirical total covariance
mean full-SG prediction covariance
PSD excess model discrepancy covariance
empirical total correlation
mean full-SG prediction correlation
PSD excess model discrepancy correlation
component-unit product
Fx
Fy
Fz
Tx
Ty
Tz


inertia_over_mass_xy_nominal: acceleration residual objective
specific acceleration [m/s^2]
angular acceleration [rad/s^2]
SG observed
model
x
y
z
time [s]


inertia_over_mass_xy_nominal: gimbal measurement audit (objective=measured_sg)
gimbal 1: RMSE=0.0759, max=0.2095 rad
gimbal 2: RMSE=0.08099, max=0.2013 rad
gimbal 3: RMSE=0.07393, max=0.172 rad
gimbal 4: RMSE=0.07131, max=0.1737 rad
raw
SG
command replay
angle [rad]
time [s]

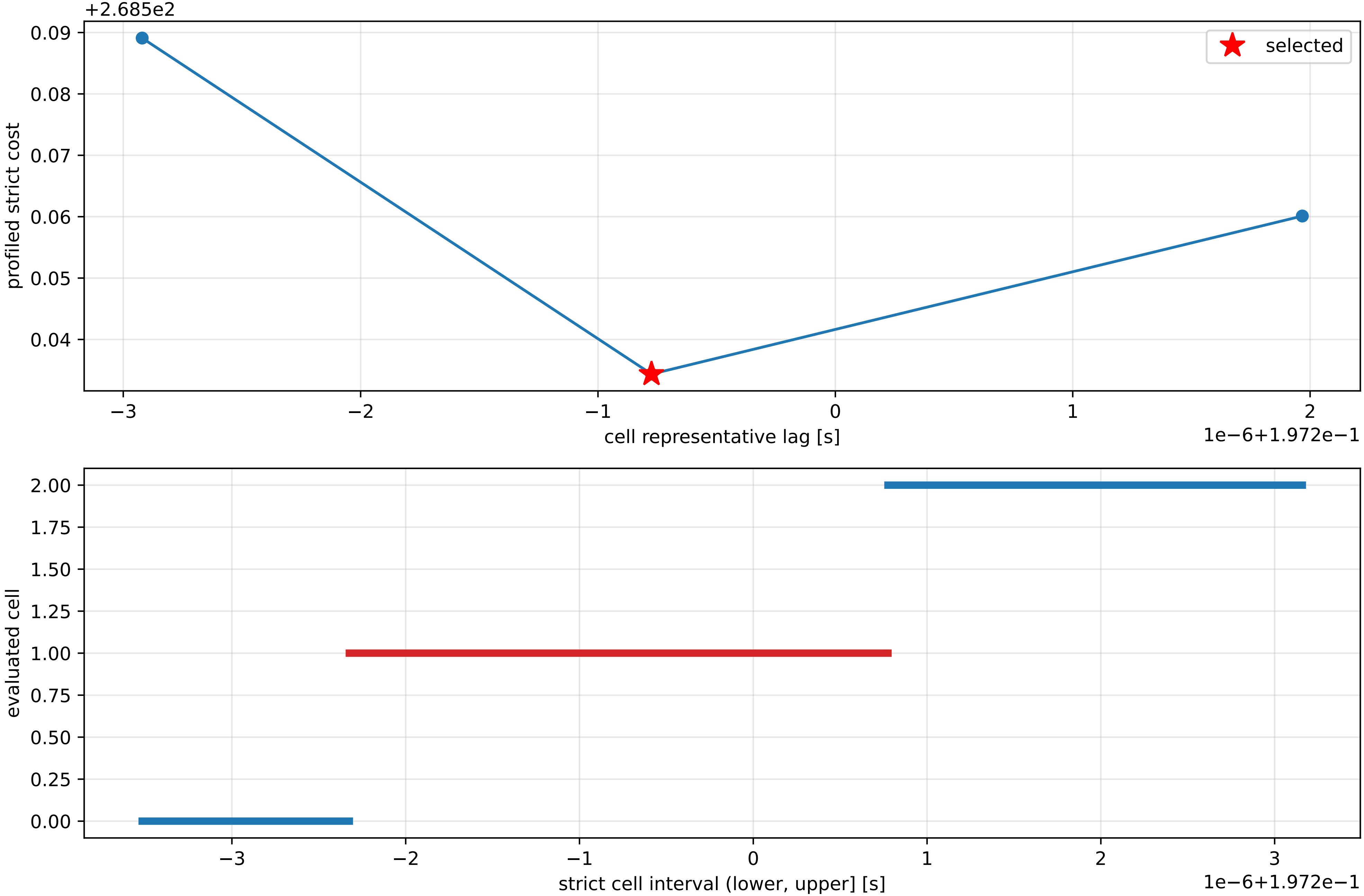


inertia_over_mass_xy_nominal: IMU comparison (diagnostic only)
gyro [rad/s]
specific force [m/s^2]
x
y
z
measured
post-fit
time [s]



inertia_over_mass_xy_nominal: smooth-to-strict continuation
rotor lag [s]
0.197325
0.197300
0.197275
0.197250
0.197225
0.197200
0.197175
objective
268.85
268.80
268.75
268.70
268.65
268.60
268.55
smooth
strict at same point
convergence error
10^{-1}
10^{-2}
|L_eps-L_0|
max thrust error
1
0.5
0.25
0.125
0.0625
0.0312
0.0156
0.00781
0.00391
0.00195
epsilon = 2^-k

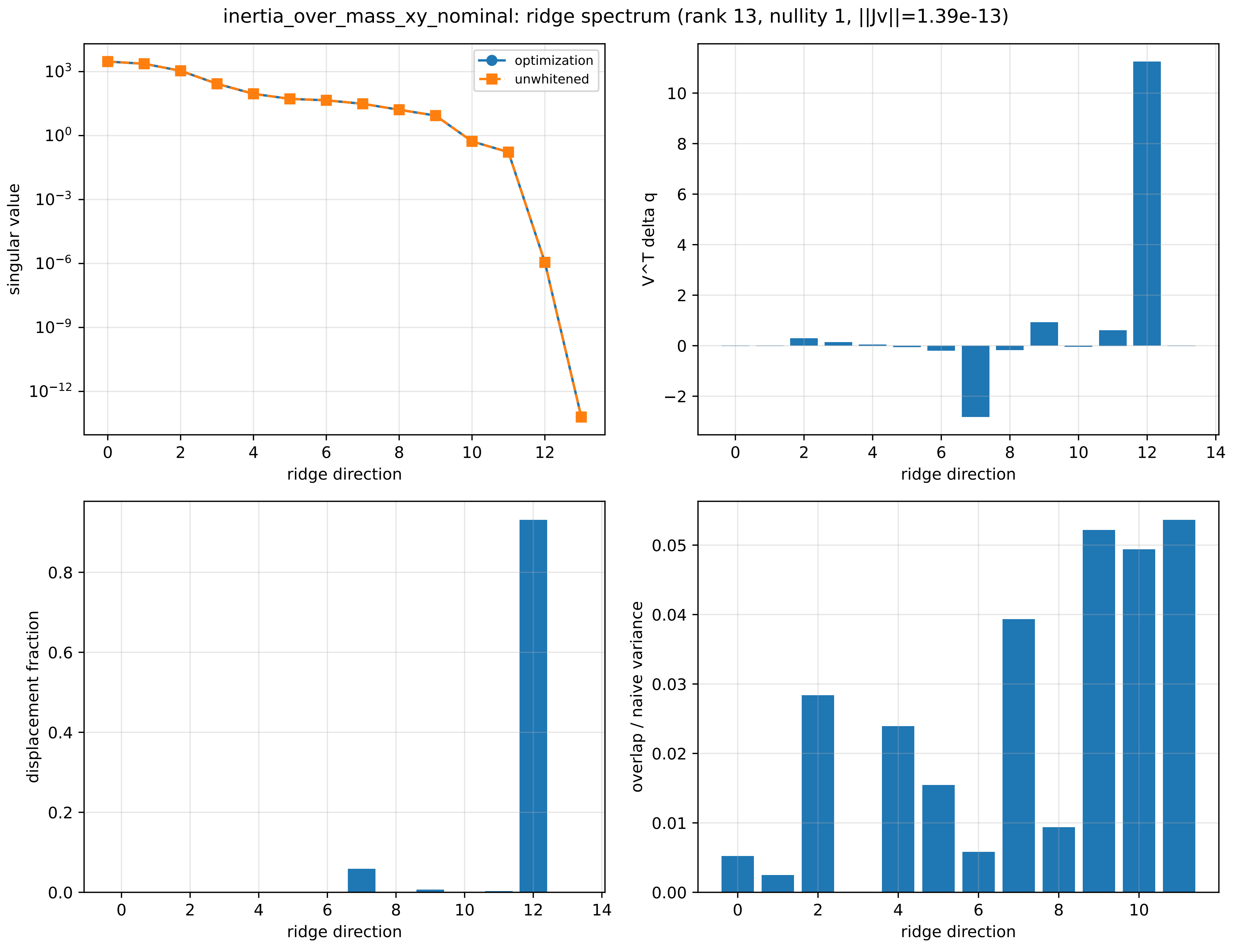
inertia_over_mass_xy_nominal: exact strict-ZOH lag cells



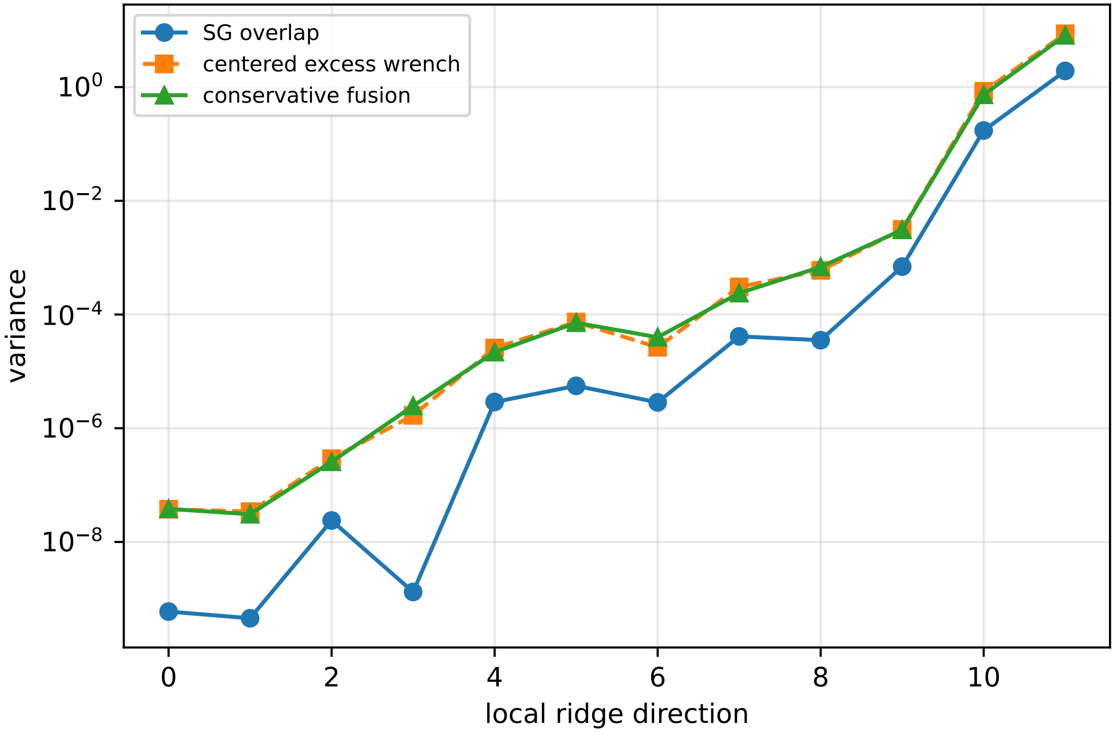
selected (0.197197676, 0.197200775] s; final smooth 0.197199575 s; data boundary=False

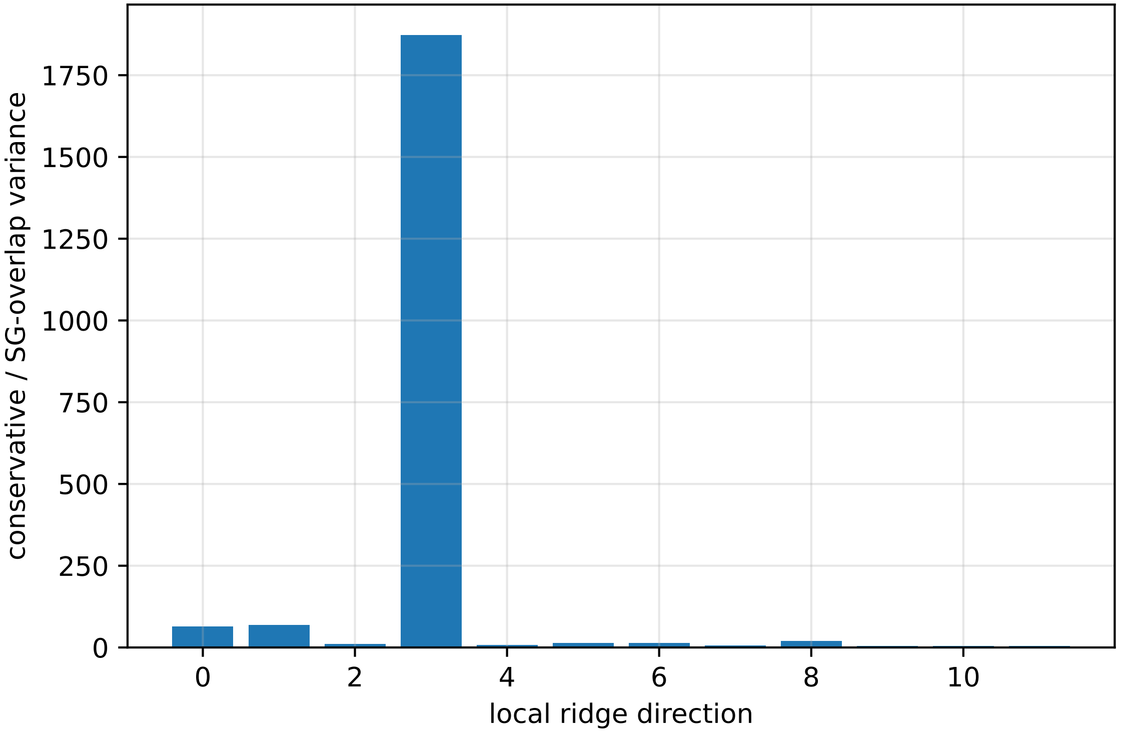
inertia_over_mass_xy_nominal: parameter estimate

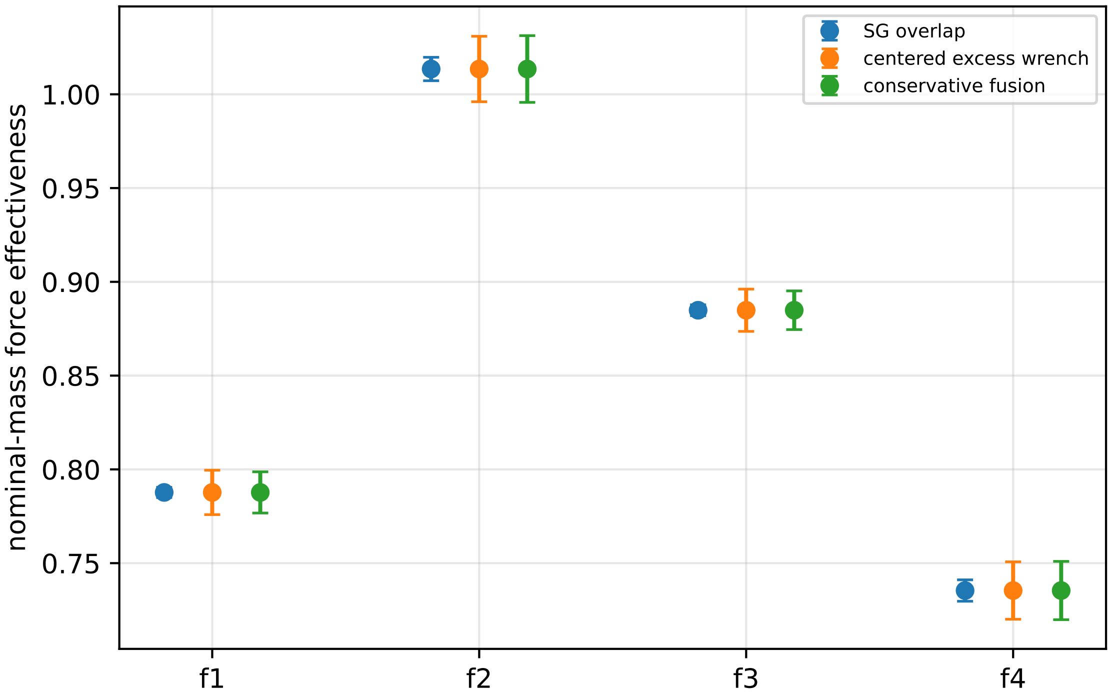
```
Case: inertia_over_mass_xy_nominal
status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 6.22353671
inertia [kg m^2]:
[[ 6.33698206e-01 -1.92703081e-06 -1.05835740e-01]
 [-1.92703081e-06  4.81171290e-01 -1.10123499e-04]
 [-1.05835740e-01 -1.10123499e-04  1.06581019e+00]]
CoG body [m]: [ 0.01745656 -0.00405043 -0.04430243]
force effectiveness: [2.08469261 2.68219959 2.34178349 1.94623546]

scale-free J/m [m^2]:
[[ 1.01822844e-01 -3.09635968e-07 -1.70057228e-02]
 [-3.09635968e-07  7.73147670e-02 -1.76946813e-05]
 [-1.70057228e-02 -1.76946813e-05  1.71254744e-01]]
scale-free f/m: [0.33496912 0.43097674 0.37627857 0.31272178]

rotor lag cell: (0.197197676, 0.197200775] s
representative: 0.197199225 s
```


inertia_over_mass_xy_nominal: ridge spectrum (rank 13, nullity 1, ||Jv||=1.39e-13)
optimization
unwhitened
singular value
ridge direction
V^T delta q
ridge direction
displacement fraction
ridge direction
overlap / naive variance
ridge direction

inertia_over_mass_xy_nominal: Conservative fusion uncertainty

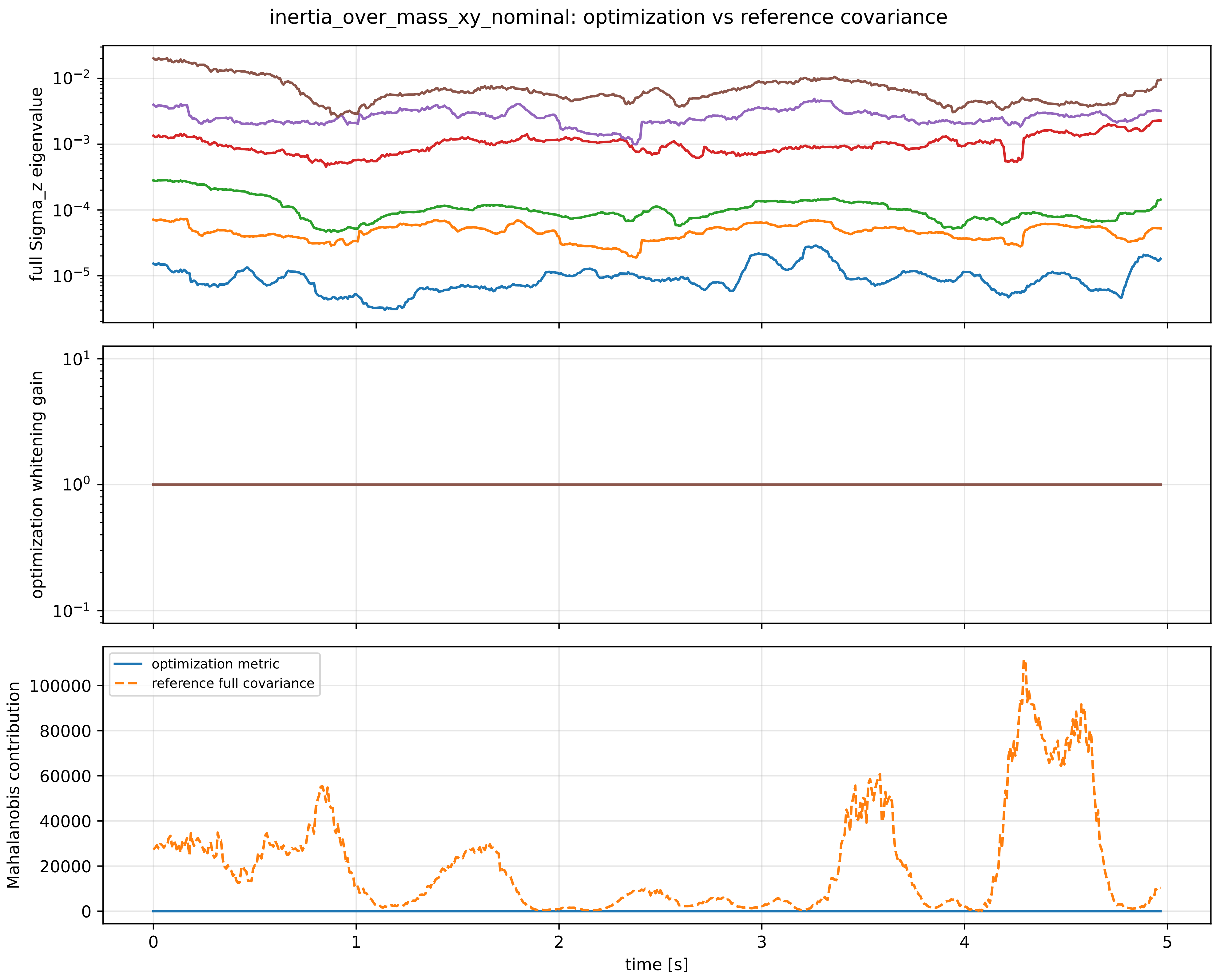


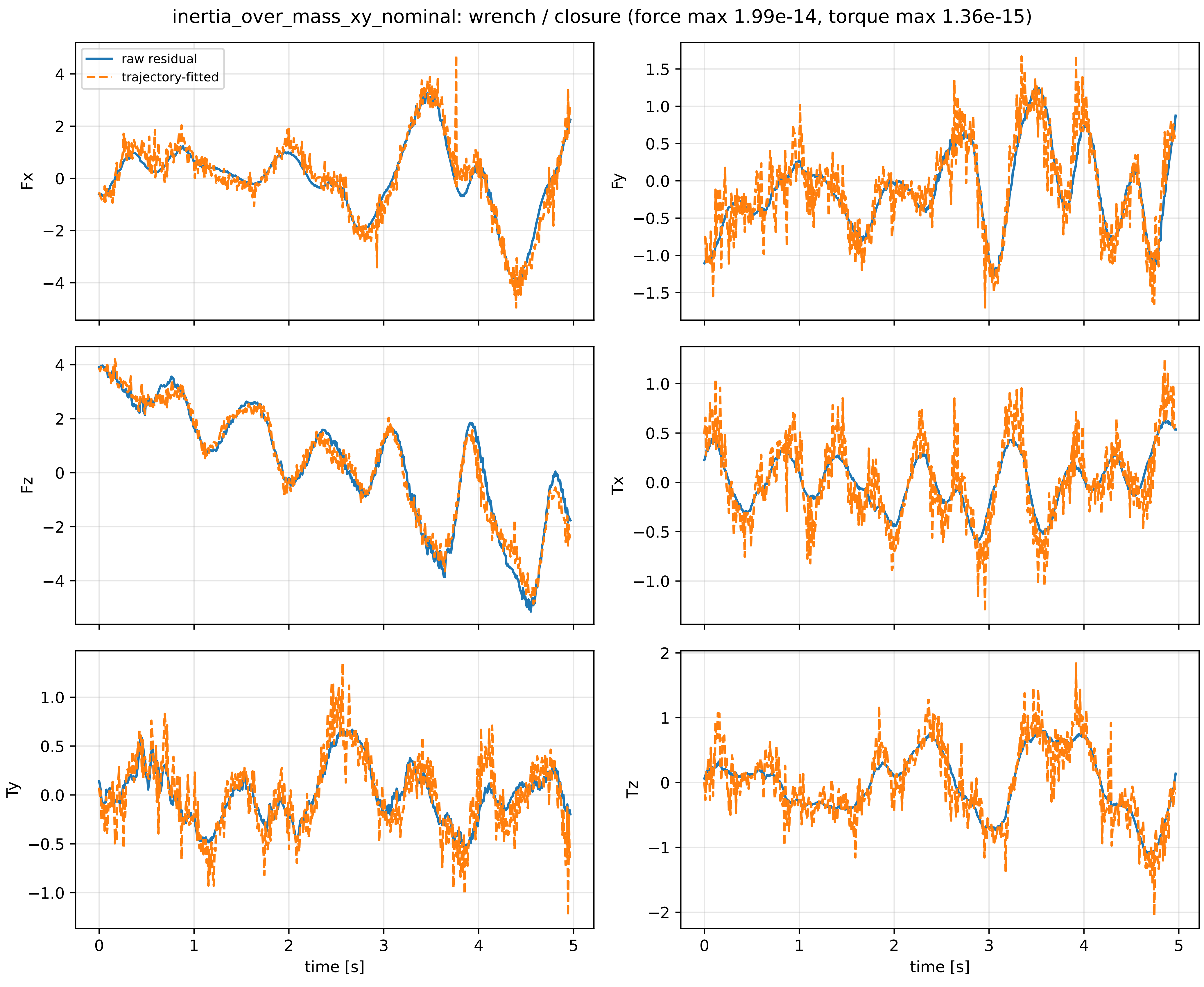





```
Top three non-gauge ambiguous directions
tr(M_res) / tr(M_SG) = 44.21
wrench-to-acceleration recovery max error = 3.19e-15
exact scale gauge ridge index = 13 (alignment 1)

index 11: variance=8.115, ratio=4.217
  log_mass=-0.0343; second_moment_diag_1=-0.425
  second_moment_diag_2=-0.033; second_moment_diag_3=+0.63
  second_moment_offdiag_12=+0.0563; second_moment_offdiag_13=-0.642
  second_moment_offdiag_23=-0.03; cog_x=+9.18e-06
  cog_y=-0.00015; cog_z=+1.52e-05
  log_force_eff_1=-0.0339; log_force_eff_2=-0.0335
  log_force_eff_3=-0.035; log_force_eff_4=-0.0347

index 10: variance=0.7327, ratio=4.248
  log_mass=+0.00221; second_moment_diag_1=+0.0252
  second_moment_diag_2=+0.00989; second_moment_diag_3=-0.0434
  second_moment_offdiag_12=+0.908; second_moment_offdiag_13=+0.0387
  second_moment_offdiag_23=-0.414; cog_x=+0.000923
  cog_y=-0.00231; cog_z=+0.000221
  log_force_eff_1=+0.00745; log_force_eff_2=+0.0139
  log_force_eff_3=-0.00276; log_force_eff_4=-0.0125

index 9: variance=0.003089, ratio=4.434
  log_mass=-0.147; second_moment_diag_1=-0.0284
  second_moment_diag_2=+0.9; second_moment_diag_3=-0.0936
  second_moment_offdiag_12=-0.00684; second_moment_offdiag_13=-0.0787
  second_moment_offdiag_23=+0.013; cog_x=+0.0219
  cog_y=-0.0136; cog_z=-0.00386
  log_force_eff_1=-0.185; log_force_eff_2=-0.022
  log_force_eff_3=-0.0936; log_force_eff_4=-0.329
```

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.


inertia_over_mass_xy_nominal: optimization vs reference covariance
full Sigma_z eigenvalue
optimization whitening gain
Mahalanobis contribution
optimization metric
reference full covariance
time [s]


inertia_over_mass_xy_nominal: wrench / closure (force max 1.99e-14, torque max 1.36e-15)
raw residual
trajectory-fitted
Fx
Fy
Fz
Tx
Ty
Tz
time [s]
time [s]
0
1
2
3
4
5

inertia_over_mass_xy_nominal: optional parameter-prior audit

```
Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.
name: pseudo_condition_inertia_over_mass_xy_nominal
role: pseudo_conditioning_ablation
source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/scalar/inertia_over_ma
SHA256: 0e9e1cf84b655f89c02f04503cdbe3c8cb0feb8b35995087c0bc407814199394

data objective:  268.534361187
prior objective: 5.20550853198e-09
total objective: 268.534361192

This configuration uses an intentionally tight artificial standard deviation
for an ablation-style pseudo-conditioning experiment; it is not a calibrated
physical uncertainty.

factor: inertia_over_mass_xy_nominal (inertia_over_mass_m2)
  components: xy
  target: [-3.09533934e-07]
  std:    [1.e-06]
  final:  [-3.09635968e-07]
  error:  [-1.02034392e-10]
  standardized residual: [-0.00010203]
  factor objective: 5.20550853198e-09
```